

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Lecture-3 Scribe Submission

Name: Jashvi Shah
Enrollment No: AU2440026
Email: jashvi.s@ahduni.edu.in
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CSE400 – Lecture 3 Scribe

Project Kickoff and Introduction to Undergraduate Research Programme (UGRP)

1. Objective of the Lecture

Lecture 3 introduces:

- The CSE400 project framework (30%)
- Team formation and execution workflow
- Milestone-based project development (M1–M6)
- Scribe and artifact expectations
- Introduction to UGRP (Undergraduate Research Programme)
- MICxN Research Lab ecosystem and outcomes

The lecture establishes how students transition from coursework to research-driven, milestone-oriented project execution.

2. CSE400 Project Structure

Weightage

The semester project carries 30% of the total course evaluation.

Team Formation

Teams must be formed by

17 January 2026 (Saturday – End of Day).

This deadline marks the start of formal project execution.

3. Project Execution Guidelines

Each group must complete six milestone submissions (M1–M6) during the semester.

Additionally, multiple artifact types are required:

- Concept Evolution Maps
- Scribe: Process & Decision Making
- Multimodal Artifacts (Video / Audio / Visual)
- Question-Driven Artifacts
- Collaboration & Team Dynamics Artifacts

Deliverables include:

- Codes
- Reports
- Videos
- Other materials as specified.

Evaluation includes:

- Team assessment (before and after mid-semester)
- Final project viva
- Final submission at semester end.

4. Step-by-Step Project Workflow (M1–M6)

The lecture defines a sequential execution pipeline.

M1 – Kickstart

Includes:

- Team formation
- Area identification
- Background study
- Motivation
- Problem formulation

This stage converts a general interest area into a clearly defined problem.

M2 – Mathematical Modeling

Students mathematically model their chosen domain using:

- Random Variables (RV)
- PMF / PDF
- CDF
- Multivariate Random Variables
- Joint PMF / PDF / CDF

Purpose: convert real-world problems into probabilistic representations.

M3 – Coding

Implementation phase involving:

- Simulation
- Computation

Models developed in M2 are translated into executable code.

M4 – Inference

Students must:

- Choose a randomized algorithm
- Understand its mechanism
- Implement it

This milestone introduces algorithmic reasoning.

M5 – Randomized Algorithms

The selected randomized algorithm is applied to the domain problem.

Students compare:

Randomized results vs Deterministic approaches

Goal: derive new inferences.

M6 – Final Analysis

Includes:

- Deriving bounds
- Performing analysis
- Compiling final deliverables

This completes the research pipeline.

5. Submission Types

Submission #1 – Concept Evolution Maps

Used to visually organize ideas.

Tools:

- Miro
- draw.io

Submission #2 – Scribe (Learning Reflection Logs)

Two types:

- Lecture Scribe
- Project Scribe

Lecture scribes:

- Assigned to two groups per lecture
- Must reflect lecture content
- Minimum 8–10 pages

Project scribes must document:

- Decision logs (Why X over Y)
- Constraints
- Alternatives considered
- Final decisions
- Evidence used
- Trade-off matrices (Cost vs Performance vs Risk)

Total: 6 scribe submissions (bi-weekly).

Submission #3 – Multimodal Artifacts

Includes:

- Think-aloud videos
- One-minute insight videos
- Project demos

Rules:

- One video per milestone
- Duration: ~10–15 minutes
- Content matters more than editing
- PPT/Google Slides may be used

6. Undergraduate Research Programme (UGRP)

UGRP promotes the idea of a T-shaped individual:

Vertical bar → depth in one discipline

Horizontal bar → ability to work across disciplines

This model emphasizes both specialization and collaboration.

7. Philosophy of UGRP

UGRP is based on:

- Multidisciplinary learning (Arts, Science, Management)
- Experiential learning
- Research-driven education

It follows a 4D model:

- Discover
- Design
- Develop
- Deliver

Focus areas:

- CS and CSE
- Data Science and Applied AI
- Modern Computer Systems (Hardware + Software)
- Networks and IoT / IoBNT / IoV

8. MICxN Research Lab Activities

Key industry projects include:

- 5G-enabled Intelligent Transportation Systems (ITS) Testbed in Gujarat
- AI-based ITS solutions
- V2X communication between OBU and RSU
- SDK testing
- Sensor data collection
- In-car dashboard development

Research directions include:

- Deep Learning crash prevention
- Deep-RL smart signaling
- mmWave beam prediction
- C-V2X and Wi-Fi coexistence
- Testbed development is conducted at Ahmedabad University.

9. Research Collaborations and Outcomes

MICxN collaborates with international universities including:

- University of Liverpool
- University of Manchester
- NTU Singapore
- Boston College
- Monash University
- Qatar University
- University of Oulu

UGRP students have:

- Published in IEEE venues
- Received Best Paper and Best UG Presentation awards
- Joined Amazon, Google, Intel, American Express
- Pursued PhDs internationally

Multiple papers were accepted at COMSNETS and NCC conferences.

10. Current MICxN Lab Scholars

Students work on:

- Advanced spectrum sensing
- O-RAN
- 6G ISAC + UAV
- Wi-Fi sensing
- Wireless networks
- Wi-Fi 8
- Antenna design

11. Conclusion

Lecture 3 establishes:

- A milestone-driven project framework
- Artifact-based assessment
- Integration of research into undergraduate learning
- Real-world impact through MICxN lab and UGRP

The session concludes with an open Q&A.